

Caio Valerio Goulart Correia

AI Specialist, Senior Data Engineer & Backend Developer

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AI Specialist, Data Engineer, and Full-Stack Developer with a robust foundation in mechanical engineering and industrial automation, enabling a unique approach to solving complex architectural problems. Specialized in LLM fine-tuning (LoRA, QLoRA, DPO), PyTorch, Hugging Face ecosystem, local LLM serving and agentic tooling (MCP, vLLM, SGLang, llama.cpp), building highly scalable data pipelines, distributed systems, and robust backend architectures using Python, Java, and Node.js. Experienced in cloud infrastructure, container orchestration (Docker/Kubernetes), and automating CI/CD workflows for enterprise-grade AI solutions.

CORE COMPETENCIES & EXPERIENCE MATRIX

Skill / Technology	Years of Experience	Level
AI Engineering & LLMs (PyTorch, Hugging Face)	2 years	Senior
LLM Fine-Tuning & Quantization (LoRA, Unsloth, vLLM)	2 years	Senior
Backend Development	9 years	Senior
Python	9 years	Senior
SQL & NoSQL	9 years (each)	Senior
RESTful APIs	6 years	Senior
Docker	6 years	Senior
Data Engineering	5 years	Senior
Node.js	5 years	Senior
Java (Spring Boot)	4 years	Mid / Senior
AWS	4 years	Mid
GraphQL	3 years	Mid
Kubernetes	3 years	Mid
OCI Cloud	2 years	Mid

PROFESSIONAL EXPERIENCE

AI Specialist

micro1 • Aug 2026 – Present

- Drive the end-to-end development, fine-tuning, evaluation, and alignment of Large Language Models (LLMs), combining engineering principles with full-stack expertise to maximize algorithmic precision, logical reasoning, and code generation quality.
- Conduct Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO/RLHF) using PyTorch, Hugging Face (PEFT, LoRA/QLoRA), Unsloth, and Axolotl; perform adversarial testing to eliminate hallucinations and logic flaws.
- Rigorously evaluate AI code generation across Python, TypeScript, Node.js, FastAPI, React, Next.js, and JavaFX, benchmarking data structure choices, dynamic execution, and complex API integrations.
- Architect local LLM serving and agent harnesses on Windows/Linux with vLLM, SGLang, and llama.cpp on dedicated NVIDIA hardware (L4 GPU / 24GB VRAM), plus MCP tool servers for retrieval/graph tools; measure latency/throughput and iterate with custom eval harnesses.
- Develop advanced prompt engineering frameworks, System Prompts, and tool-use chains to direct AI behavior, drastically improving code synthesis and enterprise IT problem-solving performance.
- Curate high-quality training datasets and synthetic data pipelines (Self-Instruct, Evol-Instruct), establishing strict automated validation for dataset integrity, schema compliance, and model alignment.

Full-Stack Developer & DevOps Engineer

ABZ Serviços • Jan 2025 – Present

- Architected and developed EmployeeHub (Portal ABZ), an internal HR / operations portal used by 200+ employees. Stack: Next.js 15 + TypeScript + Supabase (PostgreSQL/RLS) on Vercel — DP, crew management, Man Schedule, e-Social, vacation, reimbursement, and ACL; suite also includes Timesheet (PontoFlow) and Invoice ABZ.
- Led the cloud migration of legacy on-premise Windows Servers (2012-2025), achieving 80% completion with zero downtime in production environments.
- Engineered and deployed a WireGuard-based VPN infrastructure for 50+ remote employees, drastically reducing support tickets and connection failures.
- Implemented end-to-end CI/CD pipelines using GitHub Actions, Docker containerization, and administered virtualized environments across Proxmox and AWS S3.

Python Developer & Spatial Data Engineer

BRTech3D • Mar 2024 – Mar 2025

- Developed high-performance Python scripts (NumPy, SciPy, Open3D) to automate geospatial data pipelines, optimizing the processing of dense point clouds (100+ million data points).
- Implemented data routines that increased processing speed by 40% and improved data compression, yielding significant resource and time savings across dozens of projects.
- Modeled and optimized PostgreSQL databases utilizing the PostGIS extension for complex spatial queries and seamless integration via REST APIs.

Topographical Survey Operator III (Laser Scanning)

Master Engineering Services • Aug 2023 – Mar 2024

- Operated high-precision 3D laser scanning equipment (Leica) in complex industrial and archaeological environments.
- Led field campaign planning and technical documentation, ensuring millimeter-level precision and data backup redundancy for large-scale data collection.

Topographical Survey Operator II

BRTech3D • Dec 2022 – Aug 2023

- Executed field data collection using Leica TLS equipment and introduced Quality Assurance (QA) processes for outlier cleaning and multiple scan registration.

Technical Documentation & Support Specialist

SmartStore / Jackkk Connection • Mar 2021 – Dec 2022

- Produced 50+ bilingual technical manuals and system specifications for international clients.
- Provided advanced pre-sales support and detailed hardware/software architecture mapping for complex business proposals.

Training Specialist & Automation Technician

Jackkk Connection Comercial • May 2017 – Mar 2021

- Coordinated and delivered technical training for over 500 professionals in PLCs, hydraulics, and industrial automation.
- Developed educational simulations and managed B2B technical sales, building direct relationships with international suppliers.

Technical Team Lead & Specialized Support

Diesel Line Cambuí • Aug 2016 – May 2017

- Led specialized field teams in the maintenance of large-scale engines and turbines (CAT, Cummins, MAK).
- Managed English-language technical support tickets for international clients and formally documented failure analysis processes.

Apprentice Technician

Diesel Line Cambuí • Oct 2015 – Aug 2016

- Began career providing support to commercial processes, CRM management, and strict application of compliance procedures for diesel and gas systems maintenance.

RELEVANT PROJECTS

CloudSec & FinOps Compliance Auditor

Portfolio Product • 2026

- **Live:** `cloudsec-finops-auditor.vercel.app` · **Repo:** `github.com/Caiolino00/cloudsec-finops-auditor`
- Production-oriented compliance auditor that evaluates cloud architecture scenarios against CIS / SOC 2 / FinOps-style policies, returning structured risk, cost-impact, citations, and remediation steps.
- Stack: Next.js (App Router) + TypeScript + Tailwind on Vercel; hybrid RAG (BM25 + TF-IDF + Reciprocal Rank Fusion) over versioned policies/*.md; structured Gemini audits (gemini-3.8-flash → gemini-3.6-flash fallback with 503 retry); bilingual EN/PT UI; Vitest unit tests and CI evaluation pipeline.

Portal ABZ (EmployeeHub)

ABZ Serviços · Internal HR / Operations Portal · Live

- **Live:** employee-hub-kappa-flame.vercel.app · **Repo:** github.com/Caiolino00/EmployeeHub
- Built the ABZ Group corporate portal (EmployeeHub / Painei ABZ) covering DP, crew management, Man Schedule, e-Social, vacation, reimbursement, and ACL — people and offshore schedules as the operational source of truth.
- Stack: Next.js 15 + TypeScript + Supabase (PostgreSQL/RLS) on Vercel; payroll/folha derived from portal embarkation data with WK Radar sync, printable payslips, and audited reversible schedule edits.
- Integrates with the ABZ suite: identity/collaborator data feeds Timesheet (PontoFlow) sync paths; payroll/charge-rate outputs feed Invoice generation for clients.

PontoFlow (Timesheet Manager)

ABZ Group · Timesheet + mobile · Standalone (EmployeeHub-ready)

- **Repo:** github.com/Caiolino00/PontoFlow
- Delivered PontoFlow, a bilingual (pt-BR/en-GB) timesheet system for offshore crew with manager approval, field-level annotations, and monthly deadline lock enforced in UI and RLS.
- Stack: Next.js 15 + TypeScript + Supabase + Nodemailer; email/cron reminders (T-7...T); mobile companion; multi-tenant admin (invites, per-tenant SMTP).
- Built standalone with EmployeeHub-ready sync (HMAC-authenticated import/export) so collaborator identity and timesheet periods can align with the Portal ABZ suite.

Invoice ABZ (Payroll → Invoice)

ABZ Group / ABZ International · Finance automation

- **Repo:** github.com/Caiolino00/invoice_abz
- Built a multi-company Invoice engine that turns payroll/charge-rate spreadsheets into client invoices (FMS, OMEGA, Jifmar vessels) with automatic client detection and parser routing.
- Stack: FastAPI + SQLAlchemy + SQLite + OpenPyXL; preview → invoice number/date → per-person invoices or ZIP; isolated dev/prod databases.
- Closes the ABZ financial loop: Portal/folha and charge-rate data become standardized invoices for ABZ Group and ABZ International without rewriting code per client.

Model Context Protocol (MCP) / Knowledge Graph

Personal / Local AI stack

- Built MCP tool servers over a local knowledge graph (Python + SQLite/FTS indexing) so agents retrieve structured neighborhoods instead of stuffing raw context into prompts.
- Wired tool-calling loops to OpenAI-compatible local serving (vLLM / SGLang) and llama.cpp for low-latency/offline paths; custom harnesses for multi-step agents, retries, and tool-success eval.

TECHNICAL SKILLS

AI & LLM Engineering

PyTorch, Hugging Face (PEFT, LoRA/QLoRA, TRL), DeepSpeed, Unsloth, llama.cpp, vLLM, SGLang, Model Context Protocol (MCP), custom agent harnesses / tool-calling, LangChain, LlamaIndex, Triton Inference, Synthetic Data Generation.

Backend & Data Engineering

Python 3 (NumPy, SciPy), Java (Spring Boot), Node.js, Express, TypeScript, REST & GraphQL APIs, JWT Auth.

Cloud & DevOps

Docker, Kubernetes (Concepts), AWS S3, Linux (Ubuntu/CentOS), Proxmox, WireGuard VPN, Cloudflare, Vercel, Next.js.

Databases & CI/CD

PostgreSQL, PostGIS, SQLite, NoSQL, GitHub Actions (CI/CD), Git, Advanced Scripting (PowerShell/Python), Vitest.

EDUCATION

B.S. in Mechanical Engineering (7th Semester) UniFatecie • Jan 2024 – Present

Technical Diploma in Industrial Automation ETP Escola Técnica Brasil • Jan 2014 – Jan 2016

LANGUAGES

Portuguese:

Native |

English:

Fluent (Technical proficiency and conversational, C1) |

Spanish/Italian:

Advanced technical reading.